

Olanzapine/Fluoxetine Combination in Children and Adolescents With Bipolar I Depression: A Randomized, Double-Blind, Placebo-Controlled Trial

Holland C. Detke, PhD, Melissa P. DelBello, MD, John Landry, MMath, Roland W. Usher, MS

Objective: To assess the efficacy and safety of olanzapine/fluoxetine combination (OFC) for the acute treatment of bipolar depression in children and adolescents.

Method: Patients 10 to 17 years of age with bipolar I disorder (BP-I), depressed episode, baseline Children's Depression Rating Scale-Revised (CDRS-R) total score ≥ 40 , Young Mania Rating Scale (YMRS) total score ≤ 15 , and YMRS-item 1 ≤ 2 were randomized to OFC (6/25–12/50 mg/day olanzapine/fluoxetine; $n = 170$) or placebo ($n = 85$) for up to 8 weeks of double-blind treatment. The primary efficacy measure was mean change in CDRS-R using mixed-model repeated-measures methodology.

Results: Baseline-to-week-8 least-squares mean change in CDRS-R total score was greater for OFC-treated patients than for placebo-treated patients (-28.4 versus -23.4 , $p = .003$; effect size = .46), with between-group differences statistically significant at week 1 ($p = .02$) and all subsequent visits (all $p < .01$). Rates of and times to response and remission were statistically significantly greater for OFC- than for placebo-treated patients. The most frequent treatment-emergent adverse events in the OFC group were weight gain, increased appetite, and somnolence.

Mean weight gain at patient's endpoint was significantly greater for OFC- than for placebo-treated patients (4.4 kg versus 0.5 kg, $p < .001$). Treatment-emergent hyperlipidemia was very common among OFC-treated patients. Abnormal increases in hepatic analytes, prolactin, and corrected QT interval (QTc) were also common or very common but generally not clinically significant.

Conclusion: In this study, OFC was superior to placebo, and has been approved by the US Food and Drug Administration (FDA) for the acute treatment of bipolar I depression in patients 10 to 17 years of age. Benefits should be weighed against the risk of adverse events, particularly weight gain and hyperlipidemia.

Clinical trial registration information—A Study for Assessing Treatment of Patients Ages 10-17 with Bipolar Depression; <http://clinicaltrials.gov>; NCT00844857.

Key Words: olanzapine fluoxetine combination, bipolar, depression

J Am Acad Child Adolesc Psychiatry 2015;54(3):217–224.

Treatment of the depressive phase of bipolar disorder (BD) in children and adolescents remains a significant unmet medical need. Relative to those diagnosed in adulthood, patients diagnosed with BD during childhood or adolescence have significantly reduced quality of life¹ and greater risk of poorer psychiatric, psychosocial, and health-related outcomes,² including greater risk of suicidal behavior.^{3,4} Inadequate or delayed symptom control in these young patients may have serious impacts on social, emotional, and educational development that can leave this already-vulnerable population even further impaired and disadvantaged. Consequently, rapid and reliable intervention is critical.

Unfortunately, the pharmacological treatment of bipolar depression poses a major therapeutic challenge. Unopposed antidepressant therapy may potentially precipitate manic symptoms,⁵ and although there are multiple medications effective for the treatment of mania, treatment of the

depressive phase of the illness remains understudied in both children and adults, and many frequently used treatments, such as lithium, valproate, and lamotrigine, have limited or no evidence to support their use for the acute treatment of bipolar depression.⁶ At present, only 3 medications are approved for the acute treatment of bipolar depression in adults in the United States by the US Food and Drug Administration (FDA): quetiapine, lurasidone, and olanzapine/fluoxetine combination (OFC). With regard to testing these medications in children and adolescents, 2 randomized, double-blind studies of quetiapine were conducted in patients 10 to 17 years of age diagnosed with bipolar depression, but both studies failed to demonstrate a significant benefit of quetiapine versus placebo in that population.^{7,8} Pediatric trials of lurasidone are still pending.

OFC is a fixed-dose combination capsule, available in generic preparation, which contains the second-generation antipsychotic (SGA) olanzapine and the selective serotonin reuptake inhibitor (SSRI) fluoxetine. Olanzapine and fluoxetine can also be separately co-administered to achieve the same approximate proportions as the fixed-combination OFC capsule.^{9,10} Olanzapine is an effective mood stabilizer^{11–13} that appears to enhance the antidepressant action



Clinical guidance is available at the end of this article.

of fluoxetine,^{14,15} as demonstrated in adult studies of treatment-resistant depression, which showed OFC to be superior to fluoxetine alone.¹⁶⁻¹⁸ OFC has also demonstrated rapid and robust improvement of depressive symptoms in adults with bipolar I depression^{19,20} without increasing risk of switch to mania, and was superior to placebo,¹⁹ olanzapine monotherapy,¹⁹ and lamotrigine²¹ during acute treatment.

Although OFC has not previously been studied in patients <18 years of age, its components have been evaluated separately for other indications for which they are now FDA approved. Fluoxetine has been shown to be efficacious for the treatment of major depressive disorder (MDD) in patients 8 to 18 years of age^{22,23} and for obsessive-compulsive disorder in patients 7 to 17 years of age,²⁴ and is generally very well tolerated, with a safety profile similar to that observed in adults.²⁵ Olanzapine has been shown to be efficacious for the treatment of schizophrenia and manic or mixed episodes associated with BD in adolescents 13 to 17 years of age.^{26,27} The most frequent adverse events observed with olanzapine in adolescents included weight gain and somnolence,^{26,27} with adolescent patients likely to gain more weight, experience increased somnolence/sedation, and have greater increases in total cholesterol, triglycerides, low-density lipoprotein (LDL) cholesterol, prolactin, and hepatic aminotransferase levels than those observed in adults treated with olanzapine.¹⁰ As a result, US labeling for olanzapine warns that clinicians may wish to consider prescribing other medications first in adolescents.¹⁰ Although OFC in children and adolescents would be expected to carry a metabolic risk similar to that of olanzapine monotherapy, the previous lack of an approved medication for the treatment of pediatric bipolar depression supported the need to assess OFC as a possible acute treatment option despite this risk.

The objective of the present study was to assess the efficacy and safety of OFC versus placebo for the treatment of depressive episodes associated with bipolar I disorder (BP-I) in children and adolescents. This study was conducted at the request of the FDA as part of the Pediatric Research Equity Act.

METHOD

Study Patients

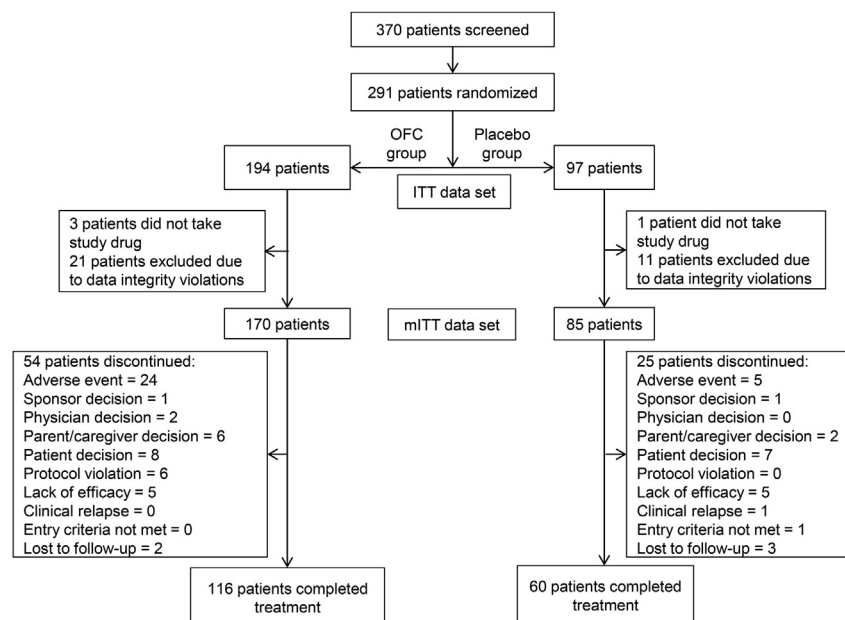
Patients were either inpatients or outpatients, 10 to 17 years of age, who met *DSM-IV-TR*²⁸ criteria for BP-I, current episode depressed, with at least 1 previous manic episode, as confirmed by the Kiddie-Schedule for Affective Disorders–Present and Lifetime (K-SADS-PL).²⁹ Patients also had to have a Children's Depression Rating Scale–Revised (CDRS-R)³⁰ total score ≥ 40 and a Young Mania Rating Scale (YMRS)³¹ total score ≤ 15 , with YMRS item 1 (Elevated Mood) score ≤ 2 at screening and randomization.

Patients were excluded who had a current or lifetime diagnosis of any of the following according to *DSM-IV-TR* criteria: schizophrenia, schizophreniform disorder, schizoaffective disorder, delusional disorder, psychotic disorder, delirium, amnestic disorder, substance-induced disorder, mental retardation, substance dependence other than nicotine or caffeine within 30 days before study entry, or a current diagnosis of autism or pervasive developmental disorder. Participants could not be actively suicidal as determined by investigator judgment.

Study Design and Treatment

This was a phase 4, randomized, double-blind, placebo-controlled study conducted between March 2009 and February 2012 at 41 centers in the United States, Mexico, and Russia. Sites were selected based on previous clinical trial experience and experience with pediatric bipolar disorder (PBD). Patients entered a 2- to 45-day screening and washout period during which all psychotropic medications were tapered and discontinued at least 25 hours before

FIGURE 1 Patient disposition. Note: ITT = intent-to-treat; mITT = modified intent-to-treat; OFC = olanzapine/fluoxetine combination.



randomization. Eligible patients were randomized to double-blind treatment with OFC or placebo for up to 8 weeks. A 2:1 randomization ratio was used to reduce potential exposure to placebo. Patients randomized to OFC were initiated at a dose of 3/25 mg (3 mg olanzapine/25 mg fluoxetine in a single capsule), which was increased to 6/25 mg at day 3, 6/50 mg at week 1, and 12/50 mg at week 2, with flexible dosing thereafter among the allowed doses of 6/25, 6/50, 12/25, and 12/50 mg. During the 2-week dose-titration period, patients could stay at the same dose or have their dose decreased if needed for safety or tolerability reasons but were discontinued if they could not tolerate a minimum dose of 6/25 mg. Patients were seen twice in the first week, weekly thereafter, and evaluated for safety/tolerability and efficacy at each visit. Patients, investigators, and all other study personnel were blinded to individual treatment assignments. Adjunctive use of benzodiazepines was permitted if ≤ 2 mg/day (lorazepam equivalents) and not within 8 hours of psychiatric assessment. Anticholinergic therapy (benztropine mesylate or biperiden ≤ 6 mg/day, or trihexyphenidyl ≤ 15 mg/day) was permitted for treatment of extrapyramidal symptoms (EPS) but not for prophylaxis. No other psychotropic drugs were permitted.

The study protocol was approved by the ethical review board at each study center, and was conducted in full accordance with ethical principles of Good Clinical Practice (GCP) and the Declaration of Helsinki and its guidelines. Informed assent and consent were obtained from all patients and their legal guardian, respectively, at study entry and before commencement of any study procedures.

Efficacy and Safety Measures

The primary efficacy measure was the mean change in CDRS-R total score from baseline to week 8. Secondary efficacy measures included the YMRS, Bipolar Depression Rating Scale (BDRS),³² and Clinical Global Impressions Scale–Bipolar Version (CGI-BP) Severity of Depression, Severity of Mania, and Overall Severity ratings.³³ Rates of and times to response and remission were also assessed. Response was defined as $\geq 50\%$ reduction from baseline to endpoint on CDRS-R total score and a YMRS item 1 score ≤ 2 at endpoint. Remission was defined as a CDRS-R total score ≤ 28 , YMRS total score ≤ 8 , and CGI-BP overall severity ≤ 3 (“mildly ill” or less) at endpoint. CDRS-R and BDRS raters had

to achieve an interclass correlation coefficient score ≥ 0.70 versus a gold-standard rating (or have no more than 4 items >1 point from the standard) before rating patients on those scales in the study.

Safety measures included assessment of adverse events (AEs), vital signs, electrocardiograms, and laboratory analytes; EPS using the Barnes Akathisia Rating Scale,³⁴ Abnormal Involuntary Movements Scale,³⁵ and Simpson-Angus Scale³⁶; and suicidality using the Columbia–Suicide Severity Rating Scale (C-SSRS).³⁷

Statistical Analyses

Analyses were based on a modified intention-to-treat (mITT) dataset (N = 255) which excluded patients who did not receive study drug (n = 4) and all patients from 2 sites that had significant data integrity violations such that data could not be considered credible or reliable (n = 32). After notification by the sponsor, both sites were audited by the FDA, who agreed with the decision to use mITT as the primary analysis set for all efficacy and safety conclusions. Analyses were also conducted using the full ITT dataset but did not reveal any significant differences in findings from the mITT dataset. All results presented represent mITT analyses unless otherwise specified.

Continuous efficacy analyses, including the primary analysis of mean change in CDRS-R, were based on mixed-model repeated-measures (MMRM) methodology using an unstructured correlation matrix with terms for baseline, country, treatment, visit, and treatment $\times$ visit interaction. Effect size was calculated post hoc using Cohen's d.³⁸ Continuous safety data were analyzed using last-observation-carried-forward (LOCF) methodology, with treatment comparisons based on analysis of covariance (ANCOVA) including terms for treatment and baseline. Mean weight change was analyzed using MMRM modeling with terms for baseline, treatment, visit, and treatment $\times$ visit interaction. Means from the MMRM and ANCOVA models represent least-squares means. Times to response and remission were assessed using Kaplan–Meier methodology and the log-rank test based on the earliest time at which patients met the respective criteria. Categorical data were analyzed using Fisher's exact test. All tests were 2-sided using a type-I error of 0.05, with no adjustments for multiple comparisons. Thus, only the primary efficacy measure comparison can be considered confirmatory; all other comparisons are considered supportive only.

TABLE 1 Baseline Patient Characteristics

Variable	OFC (n = 170)		Placebo (n = 85)		Total (N = 255)	
Age (y), mean (SD)	14.6	(2.3)	15.0	(2.1)	14.7	(2.3)
10–12 years of age, n (%)	29	(17.1)	10	(11.8)	39	(15.3)
13–17 years of age, n (%)	141	(82.9)	75	(88.2)	216	(84.7)
Sex, male, n (%)	84	(49.4)	46	(54.1)	130	(51.0)
Race, white, n (%)	119	(70.0)	61	(71.8)	180	(70.6)
Ethnicity, Hispanic, n (%)	38	(22.4)	23	(27.1)	61	(23.9)
No. of previous episodes mania, median	1		2		1	
No. of previous episodes depression, median	2		2		2	
CDRS-R total score, mean (SD)	54.6	(10.0)	53.7	(8.1)	54.3	(9.4)
YMRS total score, mean (SD)	6.1	(3.8)	6.2	(3.9)	6.1	(3.8)
CGI-BP overall severity, mean (SD)	4.4	(0.7)	4.3	(0.7)	4.4	(0.7)
Current comorbid ADHD diagnosis, n (%)	45	(26.5)	17	(20.0)	62	(24.3)

Note: ADHD = attention-deficit/hyperactivity disorder; CDRS-R = Children's Depression Rating Scale–Revised; CGI-BP = Clinical Global Impressions Scale–Bipolar Version; OFC = olanzapine/fluoxetine combination; YMRS = Young Mania Rating Scale.

RESULTS

Patient Characteristics

A total of 255 patients (170 OFC, 85 placebo) comprised the mITT population (Figure 1). Of these, 176 (69.0%) completed the study, with no statistically significant differences between treatment groups regarding study completion or reasons for discontinuation. However, discontinuation due to an AE was more than twice as high for the OFC group (14.1%) as for the placebo group (5.9%; $p = .060$). The most frequent AE leading to discontinuation for OFC-treated patients was weight gain ($n = 5$, 2.9%).

There were no statistically significant differences between groups in baseline demographics or illness characteristics (Table 1). Patients were 51% male, 71% white, 13% black, 9% mixed race, and 6% American Indian, with 24% of all patients describing their ethnicity as Hispanic. Mean age was 14.7 years. Mean symptom scores at baseline were indicative of an acutely ill patient population with moderately severe depression and an absence of mania in the current episode. Patients were predominantly outpatients; 12 were inpatients at any time during the double-blind period. Previous psychotropic medications had been taken by 61% of patients in the past year, with the most common being risperidone (14%), quetiapine (12%), aripiprazole (9%), and valproic acid (9%).

Mean doses of olanzapine and fluoxetine during the study were 7.7 mg/day (SD = 2.1) and 37.6 mg/day (SD = 8.4), respectively. The most common dose during the study was 12/50 mg, which was the target dose of the forced titration and the highest dose allowed; it was the modal dose for 39% of OFC-treated patients. Concomitant benzodiazepine use occurred more frequently in placebo-treated patients (14.1%) than OFC-treated patients (4.7%; $p = .013$). Three patients used a concomitant anticholinergic (1.2% in each group).

Efficacy

On the primary efficacy analysis, the OFC group had a 5-point greater mean decrease on the CDRS-R at week 8 than did the placebo group ($t = -3.07$, $p = .003$; Table 2). Effect size for OFC at week 8 was 0.46. The OFC group showed

statistically significantly greater improvement relative to placebo at week 1 ($p = .02$) and at all subsequent visits (all p values $< .01$) (Figure 2). Analysis using the full ITT dataset also found superiority of OFC to placebo on mean change in CDRS-R from baseline to week 8 ($p = .003$). There was no significant interaction between treatment effect and country on the CDRS-R.

More OFC-treated patients achieved response at endpoint (78.2%) than placebo-treated patients (59.2%; $p = .003$), and time to response was faster with OFC (median of ~3 weeks) than placebo (median of ~5 weeks, $p = .001$). More OFC-treated patients also achieved remission (59.0%) than placebo-treated patients (43.4%; $p = .035$), and time to remission was faster with OFC than placebo ($p = .028$). Median time to remission was ~7 weeks for OFC-treated patients; too few placebo patients achieved remission to calculate a median.

Mean changes in secondary efficacy measures are also presented in Table 2. OFC-treated patients demonstrated greater improvement than placebo on all measures of depressive symptomatology and overall illness severity (all p values < 0.05). Conversely, symptoms of mania remained low throughout the trial; the YMRS scale and CGI-BP Severity of Mania scale showed small mean decreases, with no statistically significant differences between groups. The rate of treatment-emergent mania (defined as YMRS ≥ 20 and CGI-BP Severity of Mania ≥ 5 at any time postbaseline) was also low and not statistically significantly different between groups (OFC 1.2%, placebo 0.0%, $p = 1.00$).

Safety

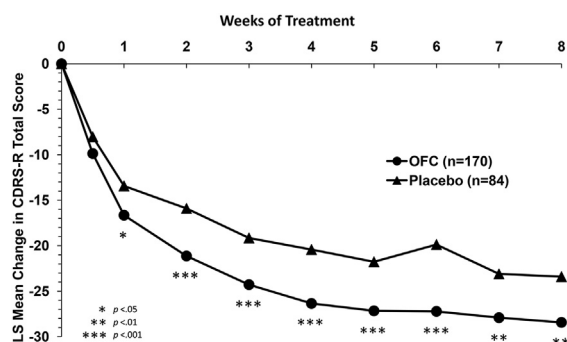
Adverse Events. Weight gain, appetite increase, somnolence, tremor, and sedation were reported statistically significantly more often by OFC- than by placebo-treated patients (Table 3). There was no statistically significant difference between groups in incidence of serious adverse events (SAEs) (OFC 5.3%, placebo 7.1%; $p = .581$), and most were related to BD. SAEs reported in >1 patient were aggression and suicidal ideation (both 1.2% in each group). SAEs occurring in 1 patient each were agitation, BD, homicidal ideation, self-injurious behavior, suicide attempt, and

TABLE 2 Changes in Efficacy Measures From Baseline to Week 8 (Mixed-Model Repeated Measures)

	OFC n = 170 Mean Baseline (SD)	OFC n = 170 LS Mean Change (SE)	Placebo n = 85 Mean Baseline (SD)	Placebo n = 85 LS Mean Change (SE)	Between-Group Difference (95% CI)	Between-Group p Value
CDRS-R total score	54.6 (10.0)	-28.4 (1.1)	53.7 (8.2)	-23.4 (1.5)	-5.0 (-8.3, -1.8)	.003
BDRS total score	25.1 (5.7)	-16.5 (0.8)	25.1 (5.4)	-13.4 (1.1)	-3.0 (-5.5, -0.6)	.016
CGI-BP overall severity	4.4 (0.7)	-2.2 (0.1)	4.3 (0.7)	-1.8 (0.2)	-0.4 (-0.7, -0.0)	.030
CGI-BP severity of depression	4.5 (0.7)	-2.4 (0.1)	4.4 (0.7)	-2.0 (0.2)	-0.4 (-0.7, -0.0)	.038
CGI-BP severity of mania	1.4 (0.7)	-0.1 (0.1)	1.4 (0.7)	-0.0 (0.1)	-0.1 (-0.3, 0.2)	.607
YMRS total score	6.1 (3.8)	-2.0 (0.5)	6.2 (3.9)	-1.6 (0.6)	-0.5 (-1.9, 1.0)	.527

Note: BDRS = Bipolar Depression Rating Scale; CDRS-R = Children's Depression Rating Scale-Revised; CGI-BP = Clinical Global Impressions Scale-Bipolar Version; LS = least-squares; OFC = olanzapine/fluoxetine combination; SE = standard error; YMRS = Young Mania Rating Scale.

FIGURE 2 Visitwise mean change in Children's Depression Rating Scale-Revised (CDRS-R) total score (mixed-model repeated measures). Note: LS = least-squares; OFC = olanzapine/fluoxetine combination.



ovulation disorder in the OFC group and depression, psychotic disorder, anaphylactic reaction, and pyoderma in the placebo group. No deaths occurred.

Weight and Vital Signs. Mean baseline-to-endpoint weight change (LOCF) was significantly greater for OFC-treated patients (4.4 kg, standard error [SE] = 0.2) than for placebo-treated patients (0.5 kg, SE = 0.3; $p < .001$). Using MMRM methodology to model expected weight gain if all patients completed the 8-week study, mean weight change from baseline to week 8 was 5.1 kg (SE = 0.3) for OFC-treated versus 0.6 kg (SE = 0.4) for placebo-treated patients ($p < .001$). Baseline-to-endpoint mean change in body mass index was also significantly greater for the OFC group (1.5 kg/m², SE = 0.1) than for the placebo group (0.1 kg/m², SE = 0.1; $p < .001$). Incidence of gaining $\geq 7\%$ of baseline body weight was significantly greater for the OFC group relative to placebo (52% versus 4%, $p < .001$), as was incidence of gaining $\geq 15\%$ of baseline body weight (14% versus 0%, $p < .001$). No placebo patients and 2 OFC-treated patients gained $\geq 25\%$ of their baseline. There were no statistically significant between-group differences in baseline-to-endpoint mean changes in blood pressure, pulse, or body temperature. Orthostatic hypotension did not occur statistically significantly more often with OFC than with placebo.

Lipids and Glucose. OFC-treated patients had mean baseline-to-endpoint increases in fasting total cholesterol (16.3 mg/dL, SE = 1.9), LDL cholesterol (9.7 mg/dL, SE = 1.5), and triglycerides (35.4 mg/dL, SE = 5.3), whereas placebo-treated patients had small mean decreases (all between-group p values $< .001$). Mean changes in fasting glucose and high-density lipoprotein (HDL) were at or near 0.0 mg/dL, with no differences between groups. Relative to placebo-treated patients, OFC-treated patients had higher rates of treatment-emergent shifts above the upper limit of normal (ULN) for fasting total cholesterol (28.9% versus 8.2%; $p < .001$), LDL cholesterol (19.7% versus 6.5%; $p = .010$), and triglycerides (52.3% versus 27.3%; $p = .003$) at any time. Treatment-emergent shifts above ULN for fasting glucose occurred in 11.0% of OFC-treated patients and 3.9% of placebo-treated patients ($p = .085$). Six OFC-treated

patients (4.8%) shifted from < 100 mg/dL fasting glucose to ≥ 126 mg/dL at any time, versus 1 patient in the placebo group (1.5%, $p = .425$).

Prolactin and Other Laboratory Values. A greater baseline-to-endpoint mean increase in prolactin occurred in the OFC group (8.7 ng/mL, SE = 0.8) compared to the placebo group (0.7 ng/mL, SE = 1.1; $p < .001$), and more OFC-treated patients experienced treatment-emergent above-normal prolactin (females > 29 ng/mL, males > 20 ng/mL) relative to placebo (58.1% versus 15.6%, $p < .001$). Five female OFC-treated patients developed AEs potentially related to prolactin elevation: galactorrhea, dysmenorrhea, and ovulation disorder (possible mittelschmerz). Regarding liver enzymes, OFC-treated patients experienced greater baseline-to-endpoint mean increases in alanine aminotransferase (ALT = 7.6 U/L [SE = 1.5]) and γ -glutamyl transferase (GGT, 2.3 U/L [SE = 0.6]) compared with the placebo group, which showed little or no change. Although a significant number of OFC- versus placebo-treated patients shifted from within normal limits at baseline to $\geq 1 \times$ ULN at any time postbaseline on ALT (48% versus 3%) and aspartate transaminase (AST, 37% versus 10%), most of these changes were transient and not clinically significant. Five OFC-treated patients had ALT shifts from normal to $\geq 3 \times$ ULN. No patients met Hy's rule (ALT $\geq 3 \times$ ULN and total bilirubin $\geq 2 \times$ ULN), and no patients developed hepatic-related symptomatology.

Electrocardiographic Findings. Mean baseline-to-endpoint increase in heart rate was greater for OFC-treated patients (4.5 beats/min, SE = 0.8) than for placebo-treated (1.0 beats/min, SE = 1.2, $p = .013$). Mean baseline-to-endpoint increase in Fridericia-corrected QT interval (QTcF) was also greater for OFC-treated patients (8.2 milliseconds, SE = 1.0) than for the placebo-treated patients (-1.1 milliseconds, SE = 1.4, $p < .001$). A total of 12.0% of OFC-treated patients experienced QTcF increase ≥ 30 milliseconds from baseline compared to 1.3% of placebo-treated patients ($p = .005$). Only 1 patient developed QTcF ≥ 460 milliseconds (maximum of 464 milliseconds), and no patients increased by ≥ 60 milliseconds from baseline at any time.

Extrapyramidal Symptoms. Mean changes on the EPS scales were near 0, with no statistically significant differences between treatment groups. Rates of treatment-emergent akathisia, parkinsonism, and dyskinesia were all $\leq 1.3\%$ in both groups and were not statistically different between groups.

Suicidality. Rates of suicidal ideation and behavior were low throughout the study, with no statistically significant difference between groups. Based on the C-SSRS, treatment-emergent suicidal ideation occurred in 9.4% of OFC-treated patients and 13.1% of placebo-treated patients, and treatment-emergent suicidal behavior occurred in 1.8% of OFC-treated patients and 2.4% of placebo-treated patients.

DISCUSSION

This was the first controlled clinical trial to evaluate OFC in a pediatric population. The study met its primary objective, as OFC was found to be statistically superior to placebo in the

TABLE 3 Treatment-Emergent Adverse Events in $\geq 5\%$ of Olanzapine/Fluoxetine Combination (OFC)–Treated Patients

	OFC n = 170		Placebo n = 85		Between-Group p Value
	n	(%)	n	(%)	
Any adverse event	125	(73.5)	49	(57.6)	.015
Weight gain	34	(20.0)	1	(1.2)	<.001
Appetite increase	28	(16.5)	1	(1.2)	<.001
Headache	27	(15.9)	12	(14.1)	.854
Somnolence	27	(15.9)	2	(2.4)	<.001
Tremor	15	(8.8)	1	(1.2)	.025
Blood triglycerides increase	12	(7.1)	2	(2.4)	.151
Fatigue	11	(6.5)	6	(7.1)	1.00
Vomiting	11	(6.5)	6	(7.1)	1.00
Sedation	11	(6.5)	0	(0.0)	.018

acute treatment of major depressive episodes associated with BP-I in patients 10 to 17 years of age, based on the mean change in the CDRS-R total score. Despite a sizable placebo effect, effect size for OFC versus placebo was still moderate.³⁹ Findings were also consistently positive across secondary efficacy measures, although results should be interpreted with caution, as statements of statistical significance did not account for multiple comparisons. OFC was also more effective than placebo in helping a very large percentage of patients achieve a clinically meaningful response and achieve this response quickly. Similarly, the majority of OFC-treated patients achieved symptomatic remission, with shorter time to remission for OFC-treated than for placebo-treated patients.

These efficacy results were similar to those of pediatric fluoxetine studies in unipolar depression.^{22,23} Mean CDRS-R scores in those unipolar studies decreased from the mid-to-high 50s by approximately 20 and 22 points in the fluoxetine groups at LOCF endpoint; placebo-treated patients decreased by a mean of approximately 10.5 and 15 points.^{22,23} By comparison, the present study found an OFC-group raw mean decrease of 25 points from baseline to LOCF endpoint versus a placebo-group decrease of 19 points, suggesting a similar drug effect. Although the placebo effect in the present study was perhaps somewhat larger than those of the 2 older unipolar fluoxetine studies, it was similar to or less than those in the more recent quetiapine studies in pediatric bipolar depression.^{7,8} Thus, it is important to acknowledge nonspecific yet potentially therapeutic aspects of pediatric clinical trials such as attention from site staff, increased parental engagement, and the behavior-modulating effects of being interviewed regularly about one's behaviors. Nevertheless, OFC still appeared to provide a clinically meaningful benefit beyond placebo. This added benefit may be particularly important when a rapid reduction in symptoms is needed.

With regard to safety, OFC-treated patients had more AEs, experienced more weight gain and more clinically significant weight gain, and had significantly higher rates of hyperlipidemia than placebo-treated patients. Elevations in

hepatic enzymes and prolactin also occurred, but these typically appeared transient and not clinically significant. The safety profile was generally similar to that of OFC in adults,^{19,40} although the magnitude of some changes in lipids, hepatic analytes, and prolactin was somewhat higher, and changes in glucose were lower than in adults. Also, OFC-treated children and adolescents demonstrated a larger (although still modest) increase in QTc than previously observed in OFC-treated adults⁴⁰ or olanzapine-monotherapy-treated adolescents.^{26,27} With the exception of the QTc finding, the present results were very similar to the safety profile of olanzapine monotherapy in adolescents.⁴¹

In terms of generalizability and clinical application of the present findings, a few cautions should be made. Because dosing in the present study was geared toward ensuring that all patients reached their maximum tolerated dose to maximize efficacy for the testing of the primary outcome measure, patients may have been titrated more quickly and exposed to higher doses than might have occurred in standard clinical practice. Despite this, most patients were able to tolerate the rapid titration to standard adult doses. Nevertheless, it is important to recognize that significant metabolic risks are associated with the use of OFC. If contemplating longer-term use, clinicians should note that longer-term use of olanzapine in adolescents has been associated with further significant weight gain and metabolic changes, as well as changes in hepatic enzymes and prolactin.⁴¹ Regarding the modest prolongation in QTc discovered in some patients in this study, it should be noted that although this cardiac finding typically may be of less concern than the metabolic risks in this population, it nevertheless could present a clinical risk to a very small percentage of pediatric patients, such as those with other risk factors for QT prolongation. Therefore, as with fluoxetine,⁹ clinicians should use OFC with caution in pediatric patients known to be particularly at risk for QT prolongation.¹⁰

Some additional limitations should be considered. Because the study did not assess children <10 years of age, results may not be generalizable to younger children. Bipolar diagnosis before the age of 10 years is less frequent, and patients in the present study were more likely to be in their teens. Also, diagnosis of PBD can be quite difficult, particularly when dependent upon retrospective reporting of a manic episode. Thus, lack of corroboration of the K-SADS-based diagnosis in the present study is a potential limitation. Lack of an active comparator may also limit interpretation of the findings; however, no other agents were approved for pediatric bipolar depression.⁴² Also, the study assessed only specific fixed-combination doses of OFC. Single-capsule, fixed-combination dosing may be more convenient for patients, may improve compliance, and may eliminate risk of patients' intentionally or unintentionally discontinuing the mood-stabilizing component while leaving the antidepressant component unopposed—an action that might increase risk of switch to mania. However, similar dosing can be achieved with available dose regimens of the individual components, and clinicians may prefer the greater flexibility of separate dosing and co-administration of the components. Such co-administration may also allow

for an easier switch to a different antipsychotic or different antidepressant if needed or desired. Results of the present study also do not address whether other antipsychotic/antidepressant combinations would be similarly effective or tolerated, but future such studies should be considered.

In conclusion, although both OFC and placebo were associated with significant clinical improvement in symptoms, OFC was found to be superior to placebo in the acute treatment of depressive episodes associated with BP-I in patients 10 to 17 years of age. Safety findings included significant weight gain and metabolic changes consistent with those historically observed in adults treated with OFC or

adolescents treated with olanzapine monotherapy. Also, a modestly higher increase in QTc was observed than has previously been seen in adults treated with OFC. However, because of the morbidity and potential mortality associated with this disorder and the high medical need for an effective treatment, OFC represents an important approved option for treating children and adolescents who are acutely ill with bipolar I depression. Clinicians will need to weigh the potential risks and benefits when considering this treatment for their individual patients. &



Clinical Guidance

- Olanzapine/fluoxetine combination (OFC) was superior to placebo in the acute treatment of bipolar I depression in patients 10 to 17 years old.
- Dosing was similar to that in adults treated with OFC, with a most common dose of 12/50 mg/day olanzapine/fluoxetine.
- Benefit–risk balance should be considered carefully. Clinically significant weight gain and hyperlipidemia were very common, and abnormal increases in hepatic analytes, prolactin, and QTc were also common or very common, although generally not clinically significant. Magnitude and frequency of some changes in lipids, hepatic analytes, prolactin, and QTc were larger than in adults treated with OFC.
- OFC is FDA approved for the acute treatment of depressive episodes associated with BP-I in patients 10 to 17 years of age and may be an important option when rapid symptom improvement is required. Olanzapine and fluoxetine also can be co-administered separately to achieve the same approximate dose as the fixed-combination OFC capsule.

Accepted December 24, 2014.

Dr. Detke is with Eli Lilly and Co., Indianapolis. Dr. DelBello is with the Division of Bipolar Disorders Research, University of Cincinnati College of Medicine, Cincinnati. Mr. Landry is with Eli Lilly Canada Inc., Toronto. Mr. Usher is a retired employee of Eli Lilly and Co., Indianapolis.

This research was funded by Eli Lilly and Co.

Mr. Landry served as the statistical expert for this research.

The authors thank the investigators in the H6P-MC-HDAX Study Group, as well as all of the patients and their families who participated in this study. The authors also thank Shannon Gardell, PhD, and Rodney Moore, PhD, of inVentiv Health Clinical (funded by Eli Lilly and Co.) for assistance in drafting/editing the manuscript.

Disclosure: Dr. Detke is an employee and minor stockholder of Eli Lilly and Co. Dr. DelBello has received grant or research support from the National Institute of Mental Health, the National Institute on Alcohol Abuse and Alcoholism, the National Institute on Drug Abuse, the National Institute of Diabetes and Digestive and Kidney Diseases, the Depressive and Bipolar Alternative Treatment Foundation, the University of Cincinnati Neuroscience Institute, Amylin Pharmaceuticals, Eli Lilly and Co., Pfizer, Otsuka, Merck, Martek, Novartis, Lundbeck, Purdue, Sunovion, and Shire. She has served as a consultant to Bracket, Guilford, Pfizer, Dey Pharma, Lundbeck, Springer, Sunovion, and Supernus. She has served on the speakers' bureau of Bristol-Myers Squibb, Merck, and Otsuka. She has received royalties from Guilford Press and honoraria for presenting at the American Academy of Child and Adolescent Psychiatry's Annual Meeting. Mr. Landry is an employee and minor stockholder of Eli Lilly and Co. Mr. Usher is a retired employee and minor stockholder of Eli Lilly and Co.

Correspondence to Holland C. Detke, PhD, Eli Lilly and Company, Lilly Corporate Center, Indianapolis, IN 46285; e-mail: detkehc@lilly.com

0890-8567/\$36.00/©2015 American Academy of Child & Adolescent Psychiatry. Published by Elsevier Inc. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>). <http://dx.doi.org/10.1016/j.jaac.2014.12.012>

REFERENCES

- Freeman AJ, Youngstrom EA, Michalak E, Siegel R, Meyers OI, Findling RL. Quality of life in pediatric bipolar disorder. *Pediatrics*. 2009;123:e446-e452.
- Perlis RH, Miyahara S, Marangell LB, *et al*. Long-term implications of early onset in bipolar disorder: data from the first 1000 participants in the Systematic Treatment Enhancement Program for Bipolar Disorder (STEP-BD). *Biol Psychiatry*. 2004;55:875-881.
- Goldstein TR, Birmaher B, Axelson D, *et al*. History of suicide attempts in pediatric bipolar disorder: factors associated with increased risk. *Bipolar Disord*. 2005;7:525-535.
- Goldstein TR, Ha W, Axelson DA, *et al*. Predictors of prospectively examined suicide attempts among youth with bipolar disorder. *Arch Gen Psychiatry*. 2012;69:1113-1122.
- Pacchiarotti I, Bond DJ, Baldessarini RJ, *et al*. The International Society for Bipolar Disorders (ISBD) task force report on antidepressant use in bipolar disorders. *Am J Psychiatry*. 2013;170:1249-1262.
- Selle V, Schalkwijk S, Vazquez GH, Baldessarini RJ. Treatments for acute bipolar depression: meta-analyses of placebo-controlled, monotherapy trials of anticonvulsants, lithium and antipsychotics. *Pharmacopsychiatry*. 2014;47:43-52.
- DelBello MP, Chang K, Welge JA, *et al*. A double-blind, placebo-controlled pilot study of quetiapine for depressed adolescents with bipolar disorder. *Bipolar Disord*. 2009;11:483-493.
- Findling RL, Pathak S, Earley WR, Liu S, DelBello MP. Efficacy and safety of extended-release quetiapine fumarate in youth with bipolar depression: an 8 week, double-blind, placebo-controlled trial. *J Child Adolesc Psychopharmacol*. 2014;24:325-335.
- Prozac [US package insert]. Indianapolis: Eli Lilly and Co; 2014.
- Zyprexa [US package insert]. Indianapolis: Eli Lilly and Co; 2013.
- Tohen M, Sanger TM, McElroy SL, *et al*. Olanzapine versus placebo in the treatment of acute mania. Olanzapine HGEH Study Group. *Am J Psychiatry*. 1999;156:702-709.
- Tohen M, Jacobs TG, Grundy SL, *et al*. Efficacy of olanzapine in acute bipolar mania: a double-blind, placebo-controlled study. The Olanzapine HGGW Study Group. *Arch Gen Psychiatry*. 2000;57:841-849.

13. Tohen M, Calabrese JR, Sachs GS, *et al.* Randomized, placebo-controlled trial of olanzapine as maintenance therapy in patients with bipolar I disorder responding to acute treatment with olanzapine. *Am J Psychiatry*. 2006;163:247-256.
14. Koch S, Perry KW, Bymaster FP. Brain region and dose effects of an olanzapine/fluoxetine combination on extracellular monoamine concentrations in the rat. *Neuropharmacology*. 2004;46:232-242.
15. Zhang W, Perry KW, Wong DT, *et al.* Synergistic effects of olanzapine and other antipsychotic agents in combination with fluoxetine on norepinephrine and dopamine release in rat prefrontal cortex. *Neuropsychopharmacology*. 2000;23:250-262.
16. Brunner E, Tohen M, Osuntokun O, Landry J, Thase ME. Efficacy and safety of olanzapine/fluoxetine combination vs fluoxetine monotherapy following successful combination therapy of treatment-resistant major depressive disorder. *Neuropsychopharmacology*. 2014;39:2549-2559.
17. Shelton RC, Tollefson GD, Tohen M, *et al.* A novel augmentation strategy for treating resistant major depression. *Am J Psychiatry*. 2001;158:131-134.
18. Thase ME, Corya SA, Osuntokun O, *et al.* A randomized, double-blind comparison of olanzapine/fluoxetine combination, olanzapine, and fluoxetine in treatment-resistant major depressive disorder. *J Clin Psychiatry*. 2007;68:224-236.
19. Tohen M, Vieta E, Calabrese J, *et al.* Efficacy of olanzapine and olanzapine-fluoxetine combination in the treatment of bipolar I depression. *Arch Gen Psychiatry*. 2003;60:1079-1088.
20. Dube S, Tollefson GD, Thase ME, *et al.* Onset of antidepressant effect of olanzapine and olanzapine/fluoxetine combination in bipolar depression. *Bipolar Disord*. 2007;9:618-627.
21. Brown EB, McElroy SL, Keck PE Jr, *et al.* A 7-week, randomized, double-blind trial of olanzapine/fluoxetine combination versus lamotrigine in the treatment of bipolar I depression. *J Clin Psychiatry*. 2006;67:1025-1033.
22. Emslie GJ, Rush AJ, Weinberg WA, *et al.* A double-blind, randomized, placebo-controlled trial of fluoxetine in children and adolescents with depression. *Arch Gen Psychiatry*. 1997;54:1031-1037.
23. Emslie GJ, Heiligenstein JH, Wagner KD, *et al.* Fluoxetine for acute treatment of depression in children and adolescents: a placebo-controlled, randomized clinical trial. *J Am Acad Child Adolesc Psychiatry*. 2002;41:1205-1215.
24. Geller DA, Hoog SL, Heiligenstein JH, *et al.* Fluoxetine treatment for obsessive-compulsive disorder in children and adolescents: a placebo-controlled clinical trial. *J Am Acad Child Adolesc Psychiatry*. 2001;40:773-779.
25. Wernicke JF. Safety and side effect profile of fluoxetine. *Expert Opin Drug Saf*. 2004;3:495-504.
26. Kryzhanovskaya L, Schulz SC, McDougall C, *et al.* Olanzapine versus placebo in adolescents with schizophrenia: a 6-week, randomized, double-blind, placebo-controlled trial. *J Am Acad Child Adolesc Psychiatry*. 2009;48:60-70.
27. Tohen M, Kryzhanovskaya L, Carlson G, *et al.* Olanzapine versus placebo in the treatment of adolescents with bipolar mania. *Am J Psychiatry*. 2007;164:1547-1556.
28. American Psychiatric Association. *Diagnostic and Statistical Manual of Mental Disorders*, fourth ed text revision ed. Washington, DC: American Psychiatric Association; 2000.
29. Kaufman J, Birmaher B, Brent D, *et al.* Schedule for Affective Disorders and Schizophrenia for School-Age Children—Present and Lifetime Version (K-SADS-PL): initial reliability and validity data. *J Am Acad Child Adolesc Psychiatry*. 1997;36:980-988.
30. Poznanski EO, Mokros HB. *Children's Depression Rating Scale—Revised: Manual*. Los Angeles: Western Psychological Services; 1996.
31. Young RC, Biggs JT, Ziegler VE, Meyer DA. A rating scale for mania: reliability, validity and sensitivity. *Br J Psychiatry*. 1978;133:429-435.
32. Berk M, Malhi GS, Cahill C, *et al.* The Bipolar Depression Rating Scale (BDRS): its development, validation and utility. *Bipolar Disord*. 2007;9:571-579.
33. Spearing MK, Post RM, Leverich GS, Brandt D, Nolen W. Modification of the Clinical Global Impressions (CGI) Scale for use in bipolar illness (BP): the CGI-BP. *Psychiatry Res*. 1997;73:159-171.
34. Barnes TR. A rating scale for drug-induced akathisia. *Br J Psychiatry*. 1989;154:672-676.
35. Guy W. *ECDEU Assessment Manual for Psychopharmacology*. Rockville, MD: U.S. Department of Health, Education, and Welfare; 1976.
36. Simpson GM, Angus JW. A rating scale for extrapyramidal side effects. *Acta Psychiatr Scand Suppl*. 1970;212:11-19.
37. Posner K, Oquendo MA, Gould M, Stanley B, Davies M. Columbia Classification Algorithm of Suicide Assessment (C-CASA): classification of suicidal events in the FDA's pediatric suicidal risk analysis of antidepressants. *Am J Psychiatry*. 2007;164:1035-1043.
38. Cohen J. *Statistical Power Analysis for the Behavioral Sciences*. New York: Academic Press; 1969.
39. Cohen J. *Statistical Power Analysis for the Behavioral Sciences*. 2nd ed. Hillsdale, NJ: Lawrence Erlbaum Associates; 1988.
40. Trivedi MH, Thase ME, Osuntokun O, *et al.* An integrated analysis of olanzapine/fluoxetine combination in clinical trials of treatment-resistant depression. *J Clin Psychiatry*. 2009;70:387-396.
41. Kryzhanovskaya LA, Robertson-Plouch CK, Xu W, Carlson JL, Merida KM, Dittmann RW. The safety of olanzapine in adolescents with schizophrenia or bipolar I disorder: a pooled analysis of 4 clinical trials. *J Clin Psychiatry*. 2009;70:247-258.
42. Tohen M, McDonnell DP, Case M, *et al.* Randomised, double-blind, placebo-controlled study of olanzapine in patients with bipolar I depression. *Br J Psychiatry*. 2012;201:376-382.